Kinematics

Building Models to Describe our World

1D, nD and Circular Motion

Outline

Kinematics in One Dimension

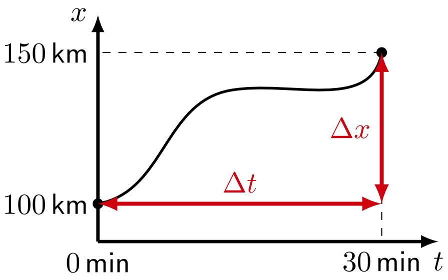
Building Models to Describe our World

1D, nD and Circular Motion

Average vs instantaneous velocity/speed

- Velocity is a measure of the rate of change of position.
- Velocity is positive(negative) if position is increasing(decreasing).
- Speed is the magnitude of velocity.
- **Average velocity**:

$$v = \frac{\Delta x}{\Delta t}$$



Position as a function of time. Average velocity is the total displacement divided by the total time.

Average vs instantaneous velocity/speed

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- Speed is the magnitude of velocity.
- **Average velocity**:

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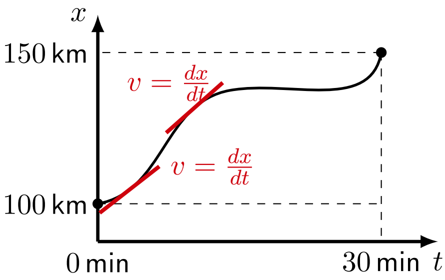
- **Instantaneous velocity** at some point in time:

$$v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$



Position as a function of time. Instantaneous velocity is the derivative at any given point.

Acceleration

- Acceleration is the rate of change of velocity
- Average acceleration :

$$a = \frac{\Delta v}{\Delta t}$$

- Instantaneous acceleration at some point in time:

$$a(t) = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$$

$$\therefore v(t) = \int a(t) dt$$

The kinematic equations for constant acceleration

- If the acceleration is constant:

$$a(t) = a$$
$$\therefore v(t) = \int a dt = at + C$$

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- The constant C can be determined from knowing that, say, at $t = 0$, $v = v_0$:

$$v(t = 0) = a(0) + C = v_0$$
$$\therefore C = v_0$$
$$\boxed{\therefore v(t) = v_0 + at}$$

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- Since velocity is the time-derivative of position, $x(t)$, position is the anti-derivative of velocity:

$$x(t) = \int v(t)dt = \int (at + v_0)dt = \frac{1}{2}at^2 + v_0 t + C'$$

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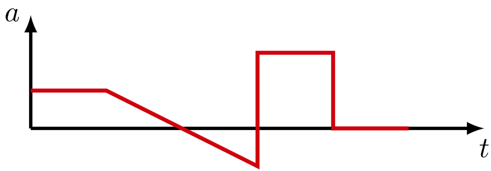
- Again, we can determine the second integration constant, $C' = x_0$, if we know that $x = x_0$ at $t = 0$:

$$\boxed{x(t) = \frac{1}{2}at^2 + v_0 t + x_0}$$

Group question

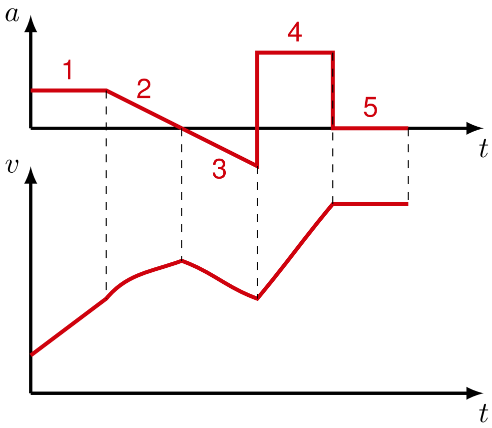
Using the following graph of $a(t)$, sketch out the corresponding $v(t)$ and $x(t)$. **Assume that, at $t = 0$, the position is negative and that the speed is positive.**



Acceleration as a function of time.

Velocity, $v(t)$

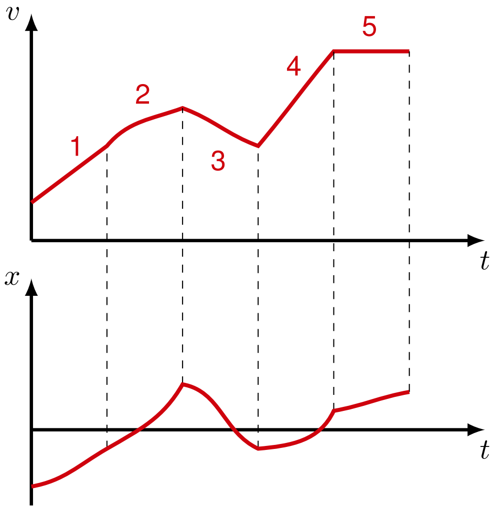
1. Acceleration is constant and positive, velocity increases linearly.
2. Acceleration is decreasing to zero, velocity increases less and less
3. Acceleration becomes more negative, velocity increases in the negative direction more and more
4. Acceleration is constant and positive, velocity increases linearly.
5. Acceleration is zero, velocity is constant.



Velocity from acceleration, the object always goes in the positive direction

Position, $x(t)$

1. Velocity increases linearly, position increase quadratically
2. Velocity increases less and less, position increases less than quadratically
3. Positive velocity decreases, position decreases less than quadratically.
4. Velocity increases linearly, position increase quadratically
5. Velocity is constant, position increases linearly



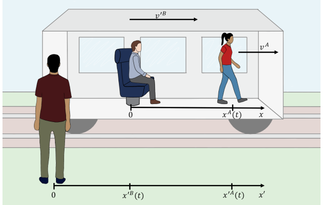
Position from velocity

1D Relative motion - notation

- We can think of different “frames of reference”, i.e. different coordinate systems that move relative to each other
- We choose to use primes (‘) to denote quantities in one frame of reference, and letters to identify the object:

$$v'^A = v'^B + v^A$$

In words: The velocity of Alice as measured in the prime reference frame (Igor, the ground) is the velocity of Brice (the train) measured relative to the ground plus **the velocity of Alice measured relative to the un-primed reference frame** (aka Brice).



Igor watches a train with Brice (sitting) and Alice (walking) in it.

Same idea with position:

$$x'^A = x'^B + x^A$$

$$x'^A = v'^B t + x^A$$

Kinematics in Multiple Dimensions

Building Models to Describe our World

1D, nD and Circular Motion

2D motion

- Same as 1D, but we need to describe both of the x and y coordinates of the object
- The most compact notation to do this is to use vectors
- Position:

$$\vec{r}(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$$

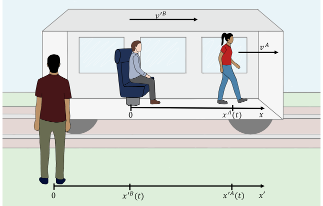
By specifying the vector, $\vec{r}(t)$, relative to the origin of an xy coordinate system, we fully specify the position of an object (namely its x and y coordinates as a function of time.

2D Relative motion - notation

- We can think of different “frames of reference”, i.e. different coordinate systems that move relative to each other
- We choose to use primes (‘) to denote quantities in one frame of reference, and letters to identify the object:

$$\vec{v}'^{A} = \vec{v}'^{B} + \vec{v}^{A}$$

In words:The velocity of Alice as measured in the prime reference frame (Igor, the ground) is the velocity of Brice (the train) measured relative to the ground plus **the velocity of Alice measured relative to the un-primed reference frame** (aka Brice).



Igor watches a train with Brice (sitting) and Alice (walking) in it.

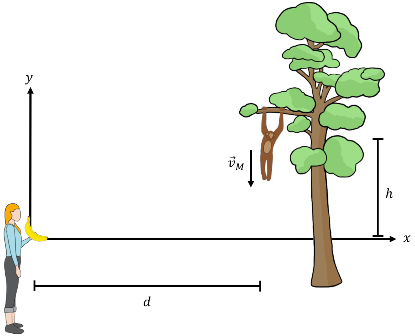
Same idea with position:

$$\vec{r}'^{A} = \vec{r}'^{B} + \vec{r}^{A}$$

$$\vec{r}'^{A} = \vec{v}'^{B} t + \vec{r}^{A}$$

Feeding the monkey: building a model

$\rightarrow$ How do we throw the banana in order for the monkey to catch it? The monkey will let go of the branch as soon as we throw the banana.



As soon as the banana is released, the monkey lets go of the branch. How to throw the banana for the monkey to catch it?

Start with initial conditions

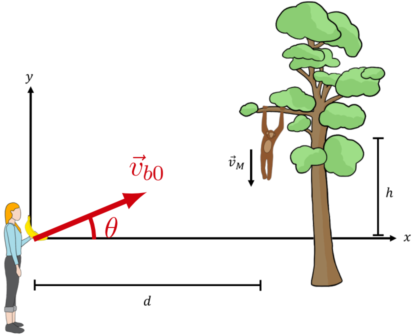
- We choose a coordinate system $\rightarrow$ in this case: origin at starting position of banana, y positive up, x positive right
- We have 2 objects, the banana and the monkey, whose motion we want to describe:

$$\vec{r}_b(t), \vec{r}_m(t)$$

- At time $t = 0$, they have initial position and velocity vectors:

$$\vec{r}_{b0}, \vec{r}_{m0}, \color{red}{\vec{v}_{b0}}, \vec{v}_{m0}$$

- We want to determine $\color{red}{\vec{v}_{b0}}$, the initial velocity of the banana



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Equate the positions of the banana and monkey

- For constant acceleration:

$$\vec{r}(t) = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a}t^2$$

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- For the banana, $\vec{r}_{b0} = 0$, this gives:

$$\vec{r}_b(t) = v_{b0}\begin{pmatrix}\cos\theta \\ \sin\theta\end{pmatrix} t + \frac{1}{2}\begin{pmatrix}0 \\ -9.8\,\mathsf{m/s^2}\end{pmatrix} t^2$$

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- For the monkey, $\vec{v}_{m0} = 0$, this gives:

$$\vec{r}_m(t) = \begin{pmatrix} d \\ h \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 \\ -9.8\,\mathrm{m/s^2} \end{pmatrix} t^2$$

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- The positions must be the same at some time t, $\vec{r}_b(t) = \vec{r}_m(t)$, writing this out for each component:

$$v_{b0}\cos\theta t = d$$
$$v_{b0}\sin\theta t - \frac{1}{2}(9.8\,\mathsf{m/s^2})t^2 = h - \frac{1}{2}(9.8\,\mathsf{m/s^2})t^2$$

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- Simplify:

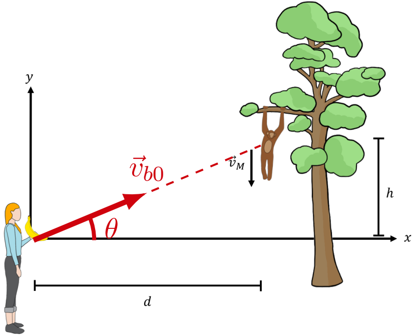
$$v_{b0}\cos\theta t = d$$
$$v_{b0}\sin\theta t = h$$

- Divide one equation by the other to eliminate t, v_{b0} and get θ:

$$\frac{\sin\theta}{\cos\theta} = \boxed{\tan\theta = \frac{h}{d}}$$

Thinking about the model, $\tan\theta = \frac{h}{d}$

- This is somewhat surprising; all we have to do is aim for the monkey...
- The speed at which we throw the banana does not matter!
- If we throw the banana faster, it will intercept the monkey higher above the ground.
- Think of this in the absence of gravity (then you obviously aim for the monkey) – in the presence of gravity, both the banana and monkey move downwards by the same amount in the same amount of time, $\frac{1}{2}gt^2$



As soon as the banana is released, the monkey lets go of the branch. How to throw the banana for the monkey to catch it? Just aim for the monkey!

Zeno's paradox

An example of Zeno's paradox: If you shoot an arrow at speed, v, towards a target a distance, D, away, it must first cover half the distance. After that, it must still go the other half, so it travels half of that remaining distance. If you think of it this way, like Zeno, you appear to have a different paradox, the arrow can never get there because there is always a small distance left. Any motion is thus impossible!

Zeno's arrow: Kinematics to the rescue!

- At constant speed, v, it will take a time, T, for the arrow to cover the distance, D:

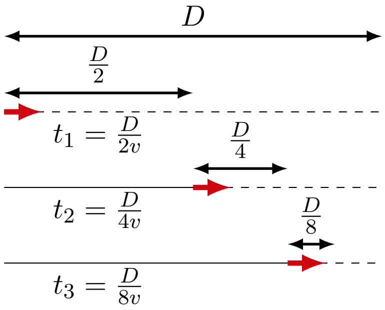
$$T = \frac{D}{v}$$

- The time that it will take to cover the first, second, third segments:

$$t_1 = \frac{D}{2v} \quad t_2 = \frac{D}{4v} \quad t_3 = \frac{D}{8v} \quad \dots \quad t_N = \frac{D}{2vN}$$

- The total time is thus also given by:

$$T = t_1 + t_2 + t_3 + \cdots = \sum_{i=1}^{N} t_i = \frac{D}{v} \sum_{1=1}^{N} \frac{1}{2i}$$



An arrow traveling Zeno's way!

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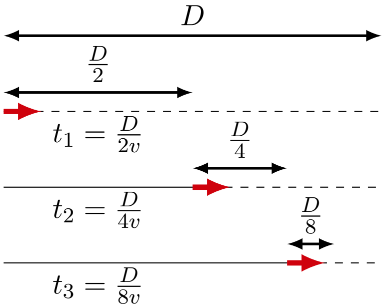

An arrow traveling Zeno's way!

Resolving the paradox:

The two ways of calculating, T, must be equivalent, thus:

$$\lim_{N \to \infty} \sum_{1=1}^{N} \frac{1}{2i} = 1$$

The geometric series converges!

Circular Motion

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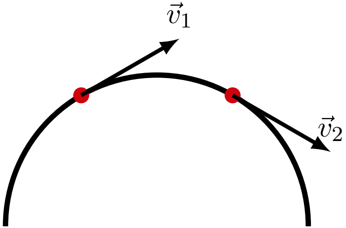
Acceleration vector: constant speed

- The acceleration vector of an object is non-zero if the velocity vector changes.

$\rightarrow$ The velocity vector can change magnitude, direction, *or both*!



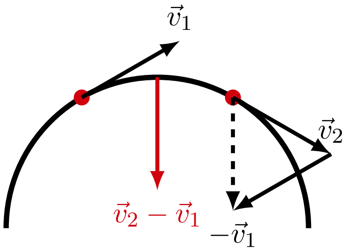
Velocity vector of an object going around a circle.

Acceleration vector: constant speed

- The acceleration vector of an object is non-zero if the velocity vector changes.

$\rightarrow$ The velocity vector can change magnitude, direction, *or both*!

- For constant speed, the acceleration vector is perpendicular to the velocity vector, in the limit of $\Delta t \to 0$ (as we'll see).



The change in velocity vector $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$ *is perpendicular to the trajectory.*

Taking the derivative of a vector

- The average acceleration vector is defined as:

$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t}$$

- Dividing a vector by a scalar (divide each component):

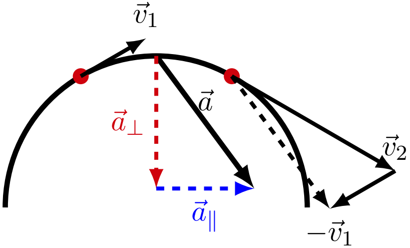
$$\begin{pmatrix} a_x \\ a_y \end{pmatrix} = \begin{pmatrix} \frac{\Delta v_x}{\Delta t} \\ \frac{\Delta v_y}{\Delta t} \end{pmatrix}$$

- The instantaneous acceleration is given by taking $\Delta t \to 0$:

$$\vec{a} = \frac{d}{dt}\vec{v} = \begin{pmatrix} a_x \\ a_y \end{pmatrix} = \begin{pmatrix} \frac{dv_x}{dt} \\ \frac{dv_y}{dt} \end{pmatrix}$$

Acceleration vector: general case

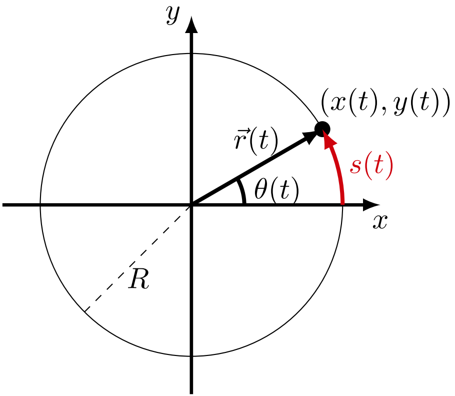
- In general, the velocity vector can change both magnitude and direction.
- The acceleration vector, $\vec{a}$, will, in general, have components that are perpendicular and parallel to the instantaneous velocity:
  - You can think of the perpendicular component, $\vec{a}_\perp$, as being responsible for the change in direction.
  - You can think of the parallel component, $\vec{a}_{||}$, as being responsible for the change in speed



The acceleration has a component perpendicular to the trajectory due to the direction changing, and a component parallel due to the magnitude changing.

Motion along a circle

- We wish to describe the motion of an object moving along a circular path.
- This is 2D motion, but constrained along a (1D) line.
- We have several ways to describe the motion:
  - Using full 2D description, $\vec{r}(t)$.
  - Using a 1D description along a curved axis: $s(t)$.
  - Using a 1D description using angular quantities: $\theta(t)$.
- They are, of course, all related and equivalent.
- 1D is usually more convenient!



Different ways to describe motion around a circle.

Angular velocity and angular acceleration

- The rate of change of the angle (angular velocity):

$$\omega = \frac{d\theta}{dt}$$

- The rate of change of the angular velocity (angular acceleration)

$$\alpha = \frac{d\omega}{dt}$$

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$$\alpha = \frac{d\omega}{dt}$$

- For **constant angular acceleration**, we have (from taking anti-derivatives):

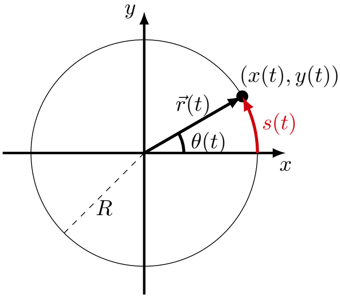
$$\omega(t) = \omega_0 + \alpha t$$
$$\theta(t) = \theta_0 + \omega t + \frac{1}{2}\alpha t^2$$

If the object was located at $\theta = \theta_0$, with angular velocity, ω_0, at time $t = 0$

Relating $\theta(t)$ and $s(t)$

- The angular position:

$$\theta(t) = \frac{1}{R} s(t)$$



Relating $\theta(t)$ and $s(t)$

- The angular position:

$$\theta(t) = \frac{1}{R}s(t)$$

- The angular velocity:

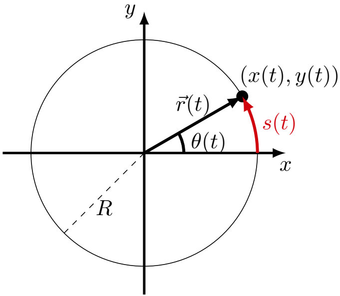
$$\omega(t) = \frac{d}{dt}\left(\frac{1}{R}s(t)\right) = \frac{1}{R}\frac{ds}{dt} = \frac{1}{R}v_s$$

- The angular acceleration:

$$\alpha(t) = \frac{d}{dt}\left(\frac{1}{R}v_s(t)\right) = \frac{1}{R}\frac{dv_s}{dt} = \frac{1}{R}a_s$$

- The “linear” or “tangential” (s, v_s, a_s) and angular (θ, ω, α) quantities are connected by the radius R:

$$\text{linear quantity} = R \times \text{angular quantity}$$



$$\boxed{s(t) = R\theta(t)}$$

$$\boxed{v_s(t) = R\omega(t)}$$

$$\boxed{a_s(t) = R\alpha(t)}$$

Vector description for circular motion

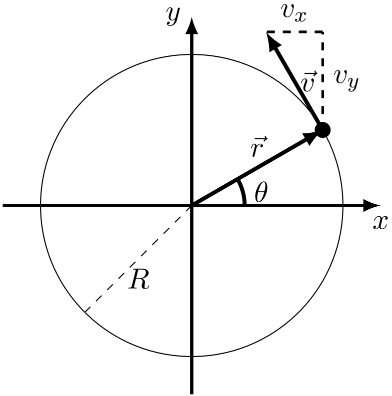
- Velocity from position:

$$\vec{r} = R\begin{pmatrix}\cos(\theta(t))\\ \sin(\theta(t))\end{pmatrix}$$

$$\therefore \vec{v} = \frac{d}{dt}\vec{r} = \frac{d}{dt}R\begin{pmatrix}\cos(\theta(t))\\ \sin(\theta(t))\end{pmatrix}$$

$$= R\begin{pmatrix}\frac{d}{dt}\cos(\theta(t))\\ \frac{d}{dt}\sin(\theta(t))\end{pmatrix} = R\begin{pmatrix}-\sin(\theta(t))\,\frac{d\theta}{dt}\\ \cos(\theta(t))\,\frac{d\theta}{dt}\end{pmatrix}$$

$$= R\omega\begin{pmatrix}-\sin(\theta(t))\\ \cos(\theta(t))\end{pmatrix}$$



Velocity and position are perpendicular. The velocity is tangent to the circle, as expected. The magnitude of the velocity vector is $v = v_s = \omega R$.

Vector description for circular motion

- The linear velocity, v_s, is the magnitude of the velocity vector:

$$\vec{v} = R\omega \begin{pmatrix} -\sin(\theta(t)) \\ \cos(\theta(t)) \end{pmatrix} = v_s(t) \begin{pmatrix} -\sin(\theta(t)) \\ \cos(\theta(t)) \end{pmatrix}$$
$$\therefore ||\vec{v}|| = v = v_s = R\omega$$

- The direction of the velocity vector is given by:

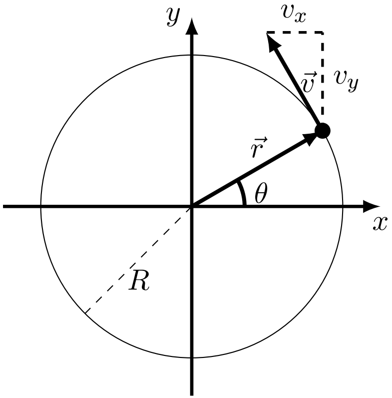
$$\hat{v} = \begin{pmatrix} -\sin(\theta(t)) \\ \cos(\theta(t)) \end{pmatrix}$$

  which is a unit vector (length of 1).

- We can write the velocity vector as:

$$\vec{v} = v(t)\hat{v}(t)$$

The direction of the velocity vector, $\hat{v}$, changes with time, and the magnitude, v, could change with time.



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- Part of the acceleration vector is due to change in speed, $\frac{dv}{dt}$, and is **parallel** to the velocity, $\hat{v}$, and part is due to change in direction, $\frac{d\hat{v}}{dt}$ and in a **different direction**:

$$\vec{a} = a_s\hat{v}(t) + v(t)\frac{d}{dt}\begin{pmatrix} -\sin(\theta(t)) \\ \cos(\theta(t)) \end{pmatrix}$$

The change in direction

- Apply the Chain Rule (again) to obtain the change in direction:

$$\frac{d\hat{v}}{dt} = \frac{d}{dt}\begin{pmatrix} -\sin\left(\theta(t)\right) \\ \cos\left(\theta(t)\right) \end{pmatrix} = \begin{pmatrix} -\cos\left(\theta(t)\right)\frac{d\theta}{dt} \\ -\sin\left(\theta(t)\right)\frac{d\theta}{dt} \end{pmatrix} = \omega\begin{pmatrix} -\cos\left(\theta(t)\right) \\ -\sin\left(\theta(t)\right) \end{pmatrix}$$

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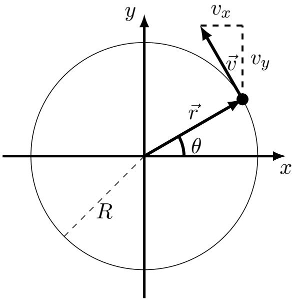
- Note that this is perpendicular to the velocity ($\hat{v}$) and has magnitude of ω. The vector $\frac{d\hat{v}}{dt}$ points towards the centre of the circle.



Finally

- The acceleration vector is given by:

$$\vec{a} = a_s \hat{v}(t) + v(t)\omega \begin{pmatrix} -\cos(\theta(t)) \\ -\sin(\theta(t)) \end{pmatrix} = \alpha(t) R \hat{v}(t) + \omega^2(t) R \begin{pmatrix} -\cos(\theta(t)) \\ -\sin(\theta(t)) \end{pmatrix}$$

where we used the relations $v(t) = \omega(t)R$ and $a_s(t) = \alpha(t)R$.
- It has two components:
  1. A component **tangent to the circle** (parallel to velocity), **if the speed changes** ("tangential" acceleration), with magnitude:

$$a_s = a_{\parallel} = a_T = \frac{dv}{dt} = R\alpha$$

  2. A component **towards the center of the circle** (perpendicular to velocity) **because the velocity changes direction** ("centripetal" or "radial" acceleration), with magnitude:

$$a_c = a_{\perp} = a_R = \omega^2 R = \frac{v^2}{R}$$